

Chapter 2

Formulas and Functions Performing Quantitative Analysis

Summary of problem:

- In this exercise you will learn how to analyze mortgage data.
- You are provided with sample mortgage data of Townsend Mortgage Company.
- This data contains loan number, house cost, down payment, mortgage rate, and the length of the loan in years.
- You will learn how to perform some basic calculations and to calculate the amount financed, the periodic interest rate, the total number of payment periods, the percent of the house cost that is financed, and the payoff year for each loan.
- In addition, you will calculate totals, averages, and other basic statistics.
- Also, you will be asked to look up interest rates from a separate table, calculate the monthly payments, and determine how much (if any) the borrower will have to pay for private mortgage insurance (PMI).

Summary:

- A workbook containing data for five mortgages financed with the Townsend Mortgage Company is provided including house cost, down payment, mortgage rate, number of years to pay off the mortgage, and the financing date for each mortgage.
- You will calculate the amount financed by each borrower by creating a formula with relative cell references that calculates the difference between the house cost and the down payment.
- After verifying the results of the amount financed by the first borrower, you will copy the formula down the Amount Financed column to calculate the other borrowers' amounts financed.

Step 1: USE A RELATIVE CELL REFERENCE

a. Open e02h1Loans

b. Click cell D8 in the Details sheet. Type = and click cell B8, the cell containing the first borrower's house cost.

c. Type - and click cell C8, the cell containing the down payment by the first borrower. Click Enter to complete the formula.

d. Double-click the cell D8 fill handle. You copied the formula down the Amount Financed column for each mortgage row.

e. Click **cell D9** and view the formula in the Formula Bar.

The formula in cell D8 is =B8-C8. The formula copied to cell D9 is =B9-C9. Because the original formula contained relative cell references, when you copy the formula down to the next row, the row numbers for the cell references change. Each result represents the amount financed for that particular borrower.

f. Press and look at the cell references in the Formula Bar to see how the references change for each formula you copied. Save the workbook.

STEP 2: USE AN ABSOLUTE CELL REFERENCE

Column E contains the mortgage rate for each loan. Because the borrowers will make monthly payments, you will modify the given annual interest rate (APR) to a monthly rate by dividing it by 12 (the number of payments in one year) for each borrower.

a. Click cell F8.

You will create a formula to calculate the monthly interest rate for the first borrower.

b. Type =E8/B5 and click Enter.

Typically, you should avoid typing values directly in formulas. Therefore, you use a reference to cell B5, where the number of payments per year is placed in the input area, so that the company can change the payment period to bimonthly (24 payments per year) or quarterly (four payments per year) without adjusting the formula.

c. Double-click the cell F8 fill handle, click cell F9, and then view the results.

An error icon displays to the left of cell F9, which displays #DIV/0!, and cell F10 displays #VALUE!. The original formula was =E8/B5. Because you copied the formula =E8/B5 down the column, the first copied formula is =E9/B6, and the second copied formula is =E10/B7. Although you want the mortgage rate cell reference (E8) to change (E9, E10, etc.) from row to row, you do not want the divisor (cell B5) to change. You need all formulas to divide by the value stored in cell B5, so you will edit the formula to make B5 an absolute reference.

d. Click Undo in the Quick Access Toolbar to undo the AutoFill process. With F8 as the active cell, click to the right of B5 in the Formula Bar.

e. Press F4 and click Enter. Excel changes the cell reference from B5 to \$B\$5, making it an absolute cell reference.

f. Double-click the fill handle to copy the formula down the Rate Per Period column. Click cell F9 and view the formula in the Formula Bar. The formula in cell F9 is =E9/\$B\$5. The reference to E9 is relative and the reference to B5 is absolute. The results of all the calculations in the Rate Per Period column are now correct.

g. Save the workbook.

STEP 3: USE A MIXED CELL REFERENCE

You will create a formula to calculate the total number of payment periods for each loan.

a. Click cell H8 and type =G8*B5.

You will multiply the number of years (25) by the number of payment periods in one year (12) using cell references.

b. Press F4 to make the B5 cell reference absolute and click Enter. You want B5 to be absolute so that the cell reference remains B5 when you copy the formula.

c. Copy the formula down the # of Pmt Periods column. The first copied formula is =G9*\$B\$5, and the result is 360.

You want to see what happens if you change the absolute reference to a mixed reference and copy the formula again. Because you are copying down a column, the column B can be relative because it will not change either way, but the row number 5 must be absolute.

d. Ensure that cell H8 is the active cell and click Undo on the Quick Access Toolbar to undo the copied formulas.

e. Click within the \$B\$5 cell reference in the Formula Bar. Press F4 to change the cell reference to a mixed cell reference: B\$5. Press Copy the formula down the # of Pmt Periods column. In this situation, using either an absolute reference or a mixed reference provides the same results.

f. Save the workbook.

Summary:

- Practice on Functions Basics
- The Townsend Mortgage Company worksheet contains an area in which you will enter summary statistics.
- In addition, you will include the current date.

STEP 1: USE THE SUM FUNCTION

The first summary statistic you calculate is the total value of the houses bought by the borrowers. You will use the SUM function.

- a.** Open e02h1Loans if you closed it at the end of Hands-On Exercise 1 and save it as e02h2Loans, changing h1 to h2.
- b.** Ensure that the Details worksheet is active and click cell B16, the cell where you will enter a formula for the total house cost.
- c.** Click AutoSum in the Editing group on the Home tab.

Excel anticipates the range of cells containing values you want to sum based on where you enter the formula—in this case, A8:D15. This is not the correct range, so you must enter the correct range.

TROUBLESHOOTING: AutoSum, like some other commands in Excel, contains two parts: the main command button and an arrow. Click the main command button when instructed to click Sum to perform the default action. Click the arrow when instructed to click the Sum arrow for additional options. If you accidentally clicked the arrow instead of Sum, press Esc to cancel the SUM function from being completed and try Step c again.

- d.** Select the range B8:B12, the cells containing house costs.

As you use the semi-selection process, Excel enters the range in the SUM function.

- e.** Click Enter.

Cell B16 contains the function = SUM(B8:B12), and the result is \$1,521,200.

- f.** Save the workbook.

STEP 2: USE THE AVERAGE FUNCTION

Before copying the functions to calculate the total down payments and amounts financed, you want to calculate the average house cost of the houses bought by the borrowers in your list.

- a.** Click the Formulas tab and click cell B17, the cell where you will display the average cost of the houses.
- b.** Click the AutoSum arrow in the Function Library group and select Average. Excel selects cell B16, which is the total cost of the houses. You need to change the range.

- c.** Select the range B8:B12, the cells containing the house costs. The function is =AVERAGE(B8:B12).
- d.** Press Enter, making cell B18 the active cell. The average house cost is \$304,240.
- e.** Save the workbook.

STEP 3: USE THE MEDIAN FUNCTION

You realize that extreme house costs may distort the average. Therefore, you decide to identify the median house cost to compare it to the average house cost.

- a.** Ensure that cell B18 is the active cell. Click Insert Function between the Name Box and the Formula Bar, or in the Function Library group on the Formulas tab. The Insert Function dialog box opens. Use this dialog box to select the MEDIAN function because it is not available on the Ribbon.
- b.** Type median in the Search for a function box and click Go. Excel displays a list of functions in the Select a function list. The MEDIAN function is selected at the top of the list; the bottom of the dialog box displays the syntax and the description.
- c.** Read the MEDIAN function description and click OK. The Function Arguments dialog box opens. It contains one required argument, Number1, representing a range of cells containing values. It has an optional argument, Number2, which you can use if you have nonadjacent ranges that contain values.
- d.** Click Collapse Dialog Box to the right of the Number1 box. You collapsed the Function Arguments dialog box so that you can select the range.
- e.** Select the range B8:B12 and click Expand Dialog Box in the Function Arguments dialog box. The Function Arguments dialog box expands, displaying B8:B12 in the Number1 box.
- f.** Click OK to accept the function arguments and close the dialog box.

Half of the houses purchased cost more than the median, \$329,750, and half of the houses cost less than this value. Notice the difference between the median and the average: The average is lower because it is affected by the lowest-priced house, \$175,500.

- g.** Save the workbook.

STEP 4: USE THE MIN, MAX, AND COUNT FUNCTIONS

You are asked to find the least and most expensive houses so typical customers of the company can be analyzed. You will use the MIN and MAX functions to obtain these statistics. In addition, you will use the COUNT function to count the number of mortgages in the sample.

- a.** Click cell B19, the cell to display the cost of the least expensive house.
- b.** Click the AutoSum arrow in the Function Library group, select Min, select the range B8:B12, and then press Enter. The MIN function identifies that the least expensive house is \$175,500.
- c.** Click cell B20. Click the AutoSum arrow in the Function Library group, select Max, select the range B8:B12, and then press Enter. The MAX function identifies that the highest-costing house is \$400,000.
- d.** Click cell B21. Type =COUNT(B8:B12) and press Enter.

As you type the letter C, Formula AutoComplete suggests functions starting with C. As you continue typing, the list of functions narrows. After you type the beginning parenthesis, Excel displays the function ScreenTip, indicating the arguments for the function. The range B8:B12 contains five cells.

- e.** Select the range B16:B21.

You want to select the range of original statistics to copy the cells all at one time to the next two columns.

- f.** Drag the fill handle to the right by two columns to copy to the range C16:D21. Click cell D21. Because you used relative cell references in the functions, the range in the function changes from =COUNT(B8:B12) to =COUNT(D8:D12).

- g.** Click cell B9, change the cell value to 425000, and then click Enter.

The results of all formulas and functions change, including the total, average, and max house costs.

- h.** Save the workbook.

STEP 5: USE THE TODAY FUNCTION

Before finalizing the worksheet you will insert the current date. You will use the TODAY function to display the current date.

- a.** Click cell B4, the cell to contain the current date.
- b.** Click Date & Time in the Function Library group, select TODAY to display the Function Arguments dialog box, and then click OK to close the dialog box. The Function Arguments dialog box opens, although no arguments are necessary for this function. Excel displays TODAY() in the Edit formula bar, and inserts the current date in Short Date format, such as 2/6/2020, based on the computer system's date.
- c.** Save the workbook. Keep the workbook open if you plan to continue with the next Hands-On Exercise.

Summary:

- Practice on Logical, Lookup, and Financial Functions
- You are asked to complete another model that might be used for future mortgage data analysis.
- As you study the model, you realize you need to incorporate logical, lookup, and financial functions.

Step 1: USE THE VLOOKUP FUNCTION

Rates vary based on the number of years to pay off the loan. You have a lookup table for three common mortgage years, and the current APR are entered. The lookup table will provide efficiency later when the rates change. You will use the VLOOKUP function to display the correct rate for each customer based on the number of years of the respective loans.

a. Open e02h2Loans and save it as e02h3Loans, changing h2 to h3.

b. Click the Payment Info worksheet tab to display the worksheet containing the data to complete. Click cell G9, the cell that will store the APR for the first customer.

c. Click the Formulas tab, click Lookup & Reference in the Function Library group, and then select VLOOKUP. The Function Arguments dialog box opens.

d. Ensure that the insertion point is in the Lookup_value box, click the Collapse Dialog Box, click cell F9 to enter F9 in the Lookup_value box, and then click the Expand Dialog Box to return to Function Arguments dialog box. Cell F9 contains the value you need to look up from the table: 25 years.

e. Press Tab, click Collapse Dialog Box to the right of the Table_array box, select the range D4:E6, and then click Expand Dialog Box to return to the Function Arguments dialog box. This is the range that contains that data for the lookup table. The Years values in the table are arranged from lowest to highest. Do not select the column labels for the range. Anticipate what will happen if you copy the formula down the column. What do you need to do to ensure that the cell references always point to the exact location of the table? If your answer is to make the table array cell references absolute, then you answered correctly.

f. Press F4 to make the range references absolute. The Table_array box now contains \$D\$4:\$E\$6.

g. Press Tab and type 2 in the Col_index_num box.

The second column of the lookup table contains the Rates that you want to return and display in the cells containing the formulas.

h. Press Tab and type False in the Range_lookup box.

To ensure an exact match to look up in the table, you enter False in the optional argument.

i. Click OK.

The VLOOKUP function uses the first loan's term in years (25) to find an exact match in the first column of the lookup table, and then returns the corresponding rate from the second column, which is 3.625%.

j. Copy the formula down the column. Spot-check the results to make sure the function returned the correct APR based on the number of years.

k. Save the workbook

STEP 3: USE THE IF FUNCTION

Lenders often want borrowers to have a 20% down payment. If borrowers do not put in 20% of the cost of the house as a down payment, they pay a **private mortgage insurance (PMI)** fee. PMI serves to protect lenders from absorbing loss if the borrower defaults on the loan, and it enables borrowers with less cash to secure a loan. The PMI fee is about 0.38% of the amount financed.

Some borrowers have to pay PMI for a few months or years until the balance owed is less than 80% of the appraised value. The worksheet contains the necessary values in the input area. You use the IF function to determine which borrowers must pay PMI and how much they will pay.

a. Click cell I9, the cell that will store the PMI, if any, for the first customer.

b. In Formulas tab, Click Logical in the Function Library group and select IF.

The Function Arguments dialog box opens. You will enter the three arguments.

c. Type E9>=\$B\$7 in the Logical_test box.

The logical test compares the down payment percentage to see if the customer's down payment is at least 20%, the threshold stored in B7, of the amount financed.

d. Press Tab and type 0 in the Value_if_true box.

If the customer makes a down payment that is at least 20% of the purchase price, the customer does not pay PMI, so a value of 0 will display whenever the logical test is true. The first customer paid 20% of the purchase price, so he or she does not have to pay PMI.

e. Press Tab and type $D9*\$B\$6/\$B\5 in the Value_if_false box.

If the logical test is false, the customer must pay PMI, which is calculated by multiplying the amount financed (D9) by the periodic PMI rate (the result of dividing the yearly PMI (B6) by the number of payments per year (B5)).

f. Click OK and copy the formula down the column.

The first, second, and fifth customers paid 20% of the purchase price, so they do not have to pay PMI. The third and fourth customers must pay PMI because their respective down payments were less than 20% of the purchase price.

g. Save the file as e02h3Loans_LastnameFirstname and submit to blackboard.

Submit e02h3Loans_LastnameFirstname on blackboard.